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The influence of haptic guidance on the production of spatio-temporal patterns

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ABSTRACT

Haptic guidance by a robot is a recent technology to support motor learning. Its mechanisms and effects are not yet well understood. One of the hypotheses is that learning of temporal characteristics is particularly susceptible to the beneficial effects of robotic guidance. In this study we investigate the influence of robotic guidance on the production of spatio-temporal patterns. Participants practiced to draw circles with the velocity profile of ellipses. Performance during the practice phase, when participants were assisted by a robotic device, as well as during the test phase, when assistance was switched off, was compared to a control group. During practice participants with robotic assistance performed better on all three dependent measures, shape, timing of the velocity modulation, and modulation amplitude. However, these differences between groups largely disappeared in the test phase. Only the difference in the amplitude of the velocity modulation remained, which was more accurate in the robot-guidance group than in the control group. This remaining difference likely results from a secondary effect of robotic guidance, namely the experience of smaller visual errors and weaker velocity modulations during practice.

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1. Introduction

During the last decade there is an increasing interest in the benefits and pitfalls of robot-assisted motor learning (for a review see Marchal-Crespo & Reinkensmeyer, 2009). Investigators from various fields study how the nervous system learns to control different types of movement and how robotic

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devices can support this learning. Most research focuses on robot-assisted motor learning of neurological patients (e.g., Kahn, Zygman, Rymer, & Reinkensmeyer, 2006; Marchal-Crespo & Reinkensmeyer, 2009; Prange, Jannink, Groothuis-Oudshoorn, Hermens, & Ijzerman, 2006; Reinkensmeyer, Emken, & Cramer, 2004; Takahashi, Der-Yeghiaian, Le, Motiwala, & Cramer, 2007), but some attention has also been given to the support of motor learning in healthy people (e.g., Emken & Reinkensmeyer, 2005; Liu, Cramer, & Reinkensmeyer, 2006; Reinkensmeyer & Patton, 2009). Here we inquire into the benefits of robot assistance for the learning of an uncommon timing of drawing movements.

Two distinct effects of robot assistance have to be distinguished in principle. The first one is its impact on performance during practice. It is rather unsurprising that robotic support during practice reduces errors (Marchal-Crespo & Reinkensmeyer, 2008; Reinkensmeyer & Patton, 2009), provides somatosensory stimulation (Marchal-Crespo & Reinkensmeyer, 2009), and leads to improved performance. These effects are beneficial in rehabilitation and in dangerous environments, where large errors are undesirable or associated with risks of accidents. In addition they can support the motivation to perform the task practiced.

The second effect of robot assistance is on learning, that is, on performance after practice when the assistance has been removed. According to Reinkensmeyer and Patton (2009), the enhanced performance during practice can be encouraging. In addition there may be non-motivational effects on motor learning. On the other hand, robotic guidance could impede learning, consistent with the majority of findings on the effects of other variants of physical guidance (Schmidt & Lee, 1999). Overall, it is still uncertain whether robot assistance provides any advantage over other training regimes at all and, if it does, which motor-skill characteristics benefit from such assistance.

The motor skills studied are mostly rather gross movement patterns which involve the arm or the leg such as walking (e.g., Emken & Reinkensmeyer, 2005) or reaching (e.g., Kahn et al., 2006; Prange et al., 2006). Fine motor skills, such as those in writing or minimally-invasive surgery, have been considered less frequently. Some studies examined the robotic support of writing or tracking movements of letters (Ben-Pazi, Ishihara, Kukke, & Ranger, 2009; Bluteau, Coquillart, Payan, & Gentaz, 2008; Henmi & Yoshikawa, 1998; Palluel-Germain et al., 2006, 2007; Solis, Avizzano, & Bergamasco, 2002; Teo, Burdet, & Lim, 2002).

Teo et al. (2002) as well as Henmi and Yoshikawa (1998) and Solis et al. (2002) developed a virtual teaching system and performed experiments to investigate its efficiency. They could show that the participants learned to write Chinese characters or calligraphy with the robotic systems. However, none of these research groups compared robot-assisted practice with other teaching methods such as traditional visual demonstrations without haptic guidance. Palluel-Germain et al. (2006,2007) made such a comparison. They developed a visuo-haptic device, Telemaque, which supported handwriting in a training program, and they compared performance before and after the training with that of a control group who practiced in a conventional way. Both in first-grade children (Palluel-Germain et al., 2006) and kindergarten children (Palluel-Germain et al., 2007) fluency in handwriting increased more after training with Telemaque than after control training.

Fluent handwriting is a skill that requires both spatial and temporal precision. There is some evidence that learning of temporal characteristics is particularly susceptible to beneficial effects of robotic guidance (Feygin, Keehner, & Tendick, 2002; Marchal-Crespo & Reinkensmeyer, 2008; Milot, Marchal-Crespo, Green, Cramer, & Reinkensmeyer, 2010), whereas acquisition of spatial characteristics may profit more from visual training (Feygin et al., 2002). However, temporal characteristics can be of different types, and not all of them may benefit from robot assistance. The present experiment was designed to determine the benefits of robot support for the learning of an unusual timing of a simple drawing movement. This is a specific temporal characteristic which is distinct, e.g., from total duration of a movement or its temporal placement relative to an environmental event. More specifically, participants had to draw a circle with the velocity profile typical for drawing an ellipse.

The difficulty of this task is due to the relation between form and velocity in drawing that is known for a long time (e.g., Derwort, 1938). According to Viviani and Terzuolo (1982), a strict relation exists in that curvature and velocity are inversely related so that with increasing curvature velocity decreases and vice versa. The equation which describes this relation is called the 2/3 power law (Viviani & Terzuolo, 1982). According to this law, and according to experimental observations, a circle is drawn with a constant velocity because its curvature is constant, whereas an ellipse is drawn with a

modulated velocity because curvature varies along its circumference. According to several studies it is very difficult or even impossible to produce movements that violate the 2/3 power law (e.g., Viviani, Campadelli, & Mounoud, 1987; Viviani & Mounoud, 1990), but Beets, Rösler, and Fiehler (2010) recently showed that, in principle, one can learn circular arm movements with an elliptic velocity profile.

Beets, Rösler, and Fiehler (2010) supported their participants by means of a motor-driven crank which enforced a circular path with velocity modulations appropriate for an elliptic path. As noted above, showing that a certain motor skill can be learned with robot support during practice does not imply that robot support has any advantages above other procedures. Therefore we compared the effects of robot-assisted practice with the effects of a non-assisted, but otherwise identical practice regime. The participants practiced to draw circles with the velocity profile of a horizontal ellipse, that is, the resultant velocity was higher at the top and bottom of the circle than at both sides. Since the timing and shape of drawing movements are tightly interrelated, we also assessed the spatial characteristics. During practice participants tracked a moving target either without or with the support of a robot. The moving target provided sort of visual guidance, the robot additional haptic guidance. During tests the velocity-modulated circles had to be drawn without the visible target and without robot assistance.

2. Methods

2.1. Participants

A total of 26 females and 12 males participated in the study. Their age ranged from 18 to 31 years (M = 24.2 years, SD = 3.10 years). 19 of them practiced with robot support in the experimental group (16 female, 3 male; mean age: 23.5 years), 19 without robot support in the control group (10 female, 9 male; mean age: 24.9 years). All participants were right-handed, had normal or corrected to normal vision and no significant language, motor or neurological dysfunction according to their own declaration. Each participant signed written informed consent and was paid 15 Euro for participation.

2.2. Apparatus

Participants drew with a pen on a digitizer (Wacom Intuos4) which sampled data at 60 Hz. The tip of the pen was attached to the end of a robot arm (Phantom Premium 1.5, SensAble Technologies), as shown in Fig. 1a. The connecting link at the end of the robot arm carried a hole into which the tip of the pen was inserted. The pen was held there by a minute rubber ring inside the walls of the hole, so that it could be tilted freely in all directions on the tablet. Only with respect to horizontal movements the link between robot arm and tip of the pen was rigid. To avoid injuries in case the Phantom got out of control, a safety barrier was placed between Phantom and participant. Fig. 1b shows the whole experimental setup with Phantom, the digitizer and the monitor (liyama S902JT VisionMaster Pro 451) which was placed at about 90 cm distance from the participant's eyes.

2.3. Design and procedure

Participants were seated in front of the digitizer in a quiet room, with walls covered by black curtains. They were instructed to watch the screen of the monitor. The room light was switched off during the experiment to prevent participants from watching their hand and the pen. The whole experiment took about one and a half hour. First the instructions were read and two practice trials were performed to allow the participants to familiarize themselves with the apparatus and the task.

Familiarization was followed by five blocks of trials. Each block consisted of a practice phase of 25 trials followed by a test phase of 15 trials. In each trial five circles had to be drawn. In the practice phase participants performed a tracking task. The position of the tip of the pen was mapped on the position of a cursor, a red filled circle of 5 mm diameter. With this cursor participants had to follow the target, a blue filled circle of 5 mm diameter. The target moved on a circular path of 10 cm

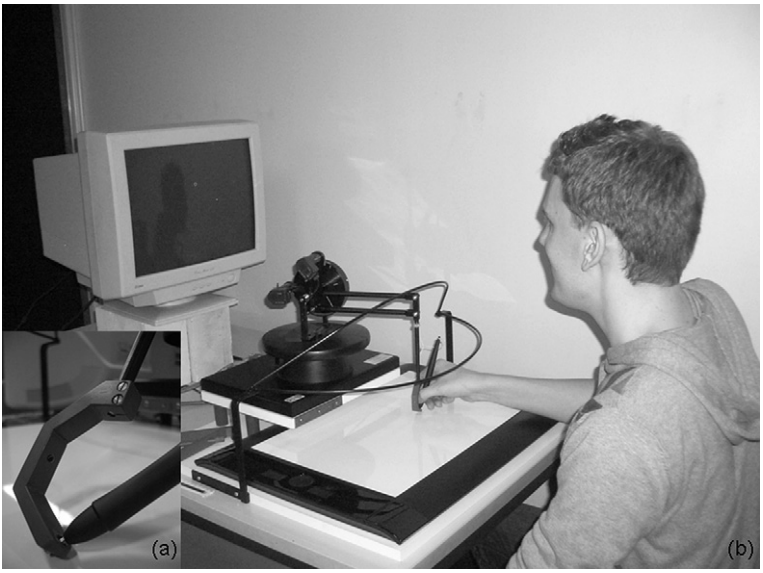

Fig. 1. Apparatus: (a) The tip of the pen was attached to the lower end of the stylus of a Phantom Premium 1.5. (b) Participants were seated in front of a digitizer tablet and faced a monitor.

diameter. Its total movement time for this circle was 3.3 s. The resultant velocity of the target was that appropriate for a horizontal ellipse with a proportion of its vertical to its horizontal axis of 0.5. Thus, velocity was higher at the top and bottom of the circular path (maximum velocity: 122 mm/s) and lower at its left and right side (minimum velocity: 61 mm/s). Neither the target nor the cursor left a visible trace on the screen. The motion of the target was sufficiently slow not to give rise to the perception of an elliptical path, as reported, e.g., by Viviani and Stucchi (1989) who used a movement time of 0.6 rather than 3.3 s.

Participants began each trial at a start position at the extreme right side of the circular path. After the cursor had been aligned with the target for two seconds, the target started to move, and the participants tracked it with the cursor for a total of five circles. During the practice trials the experimental group was assisted by the Phantom. More specifically, at each point in time there was a force field which drove the tip of the pen to its target position at which the cursor was aligned with the target on the monitor. The force was $0.3d$ N, with d as the distance of the current position of the tip of the pen from its target position (upper limit: 10 Newton). Pilot studies had revealed that this force was appropriate to minimize the tracking error in the experimental group. The participants of the control condition also drew with the pen attached to the Phantom device, but there was no haptic guidance as in the experimental group.

In the trials of each test phase participants were required to reproduce the circular trajectory with the elliptic velocity profile. At the beginning of each test trial the target was presented at the rightmost position of the circular path. After participants had aligned the cursor with the target for two seconds, the target disappeared and participants could start their movements. The cursor remained visible but left no trace on the screen. Participants were told to draw five circles and to reproduce the shape and the velocity profile as accurately as possible. After five circles had been drawn, a beep occurred, the target appeared again, and the next trial began.

2.4. Data analysis

The position-time curves of the tip of the pen were low-pass filtered (fourth-order Butterworth, 5 Hz, dual-pass). From the velocities along both dimensions of the plane, which were calculated by a central-difference algorithm, the resultant velocity was computed.

The trajectory of each trial was segmented into individual cycles by a spatial criterion. A new cycle began whenever the horizontal axis through the start position was passed. For each of the three central cycles of each trial the circularity of the trajectories and the modulation of the resultant velocity were evaluated by means of three dependent variables, a circularity index, a measure of modulation timing of velocity and a measure of modulation amplitude.

The circularity index served to assess deviations from a circular path toward an ellipse. It was determined in the following way. First, a surrounding square was fitted to the recorded path of each cycle. The center of the square served as the midpoint of the roughly circular trajectory. Second, a coordinate system was positioned in this configuration such that its origin was in the midpoint, and its abscissa was aligned with the longest diameter of the trajectory. Third, the radius was determined for each sample point, that is, its distance from the origin of the coordinate system. Fourth, the approximate circle circumscribed by the trajectory was split into four sectors with the diagonals as boundaries. The radii were averaged across the data points of each sector. For a circular path these four means are identical, but for an ellipse the means of the horizontal sectors are larger than the means of the vertical sectors. Fifth, the circularity index was defined as the ratio of the means of the horizontal sectors and the vertical sectors. It is 1 for perfect circularity, and larger values indicate elliptical deviations from circularity (Fig. 2).

The measure of modulation timing of velocity was a cross-correlation. The resultant velocity for each cycle was normalized to a length of 128 data points (with linear interpolation). In addition it was shifted such that mean velocity corresponded to zero. This normalization does not affect the dependent measures, but makes visual inspection and comparison of plotted velocity profiles easier. The target velocity was treated in the same way. Modulation timing was defined as the maximum of the cross-correlation function (Fig. 3). This measure is 1 for perfect temporal regularity of velocity modulation, and smaller than 1 otherwise, that is, when the relative temporal positions of the minima and maxima are not as required. Note that the measure is insensitive to temporal shifts of the whole profile within each cycle, so that maximal resultant velocities that are not exactly at the top and bottom of the circle. In addition the measure is insensitive to variations of scale factors. Therefore it was complemented by the measure of modulation amplitude.

The measure of modulation amplitude of velocity was derived from the peak-to-peak amplitude of the resultant velocity of each cycle, more precisely, from the difference between the means of the two peaks and the two troughs of the profile (Fig. 3). The ideal peak-to-peak amplitude, which was 61 mm/s, was subtracted from the observed one. Thus the measure of modulation amplitude is an error measure which is zero when the amplitude is exactly as required. It is positive for a too strong modulation and negative for a too weak one.

The circularity index and the measures of modulation timing and amplitude were averaged across the middle three cycles of each trial, and these means were averaged across the trials of each block. These individual means were entered in the statistical analyses. Separately for the practice phases

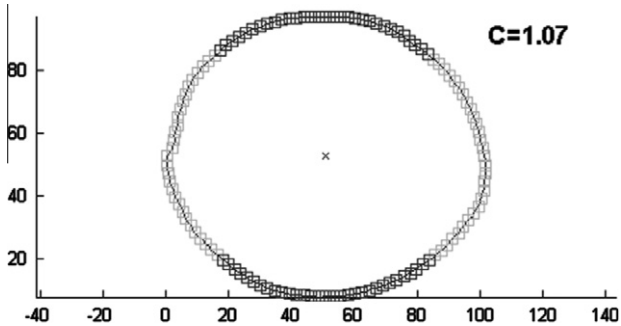

Fig. 2. Shape sample. The average radius of the horizontal segments (light gray) was divided by the average radius of the vertical segments (black), so that 1 represents perfect circularity. The circularity C = 1.07 in this example characterizes a rather elliptic form.

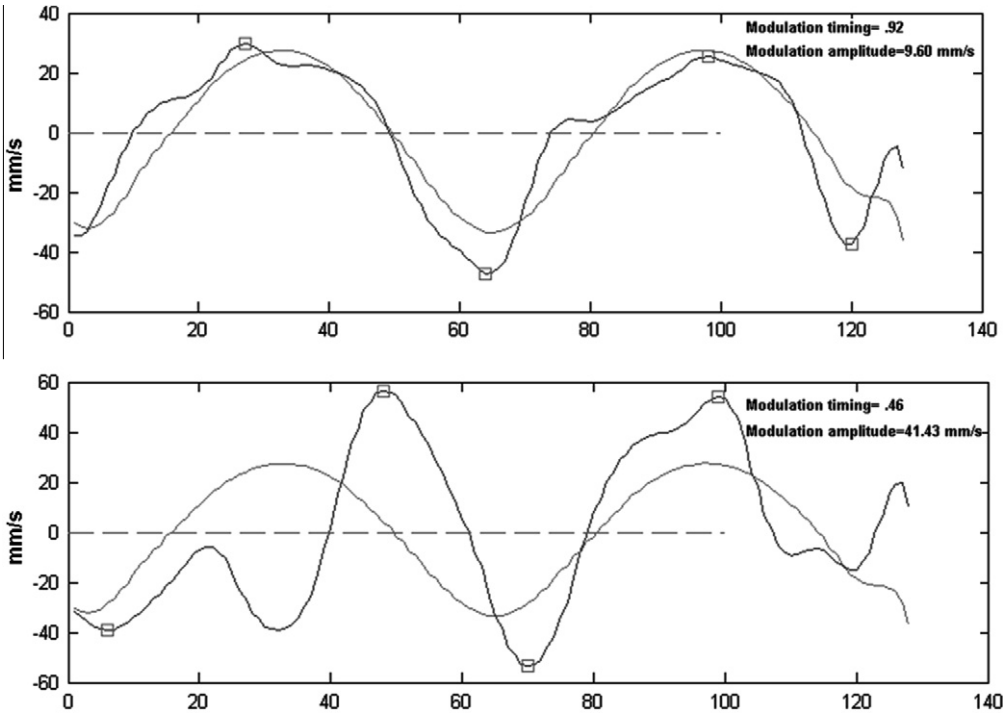

Fig. 3. Two samples of velocity curves. The curves marked by the four squares at the peaks and troughs are the observed velocity profiles, the other curves are the ideal velocity profiles. The precision of modulation timing of resultant velocity is measured by the maximum value of the cross-correlation function. The precision of modulation was measured by the deviation of the peak-to-peak amplitude from the correct amplitude; this error measure is called modulation amplitude for short. The participant of the upper graph succeeded in producing the desired velocity profile and reached a very high cross-correlation value. The modulation amplitude only deviated marginally from the desired amplitude. In contrast, the participant of the lower figure only reached a moderate cross-correlation value and exaggerated the modulation amplitude.

and test phases of the five blocks of trials and the three dependent variables two-way ANOVAs were run with the between-participant factor practice group and the within-participant factor block. Greenhouse-Geisser corrections were applied when appropriate, but the uncorrected degrees of freedom are reported together with the Greenhouse-Geisser epsilon. Effect sizes are given as partial eta-squared.

3. Results

The results of the experiment will be reported first for the practice trials and thereafter for the test trials. Whereas the practice trials reflect the immediate performance effects of the robot assistance, the test trials reflect the effects on learning.

3.1. Practice phase

As expected, the experimental group with robot assistance performed better with respect to both the circularity of the trajectory and the modulation timing and amplitude of resultant velocity (Fig. 4). In the experimental group the mean circularity index was $M = 1.017$ ($SE = .002$), whereas in the control group it was $M = 1.042$ ($SE = .002$), indicating a somewhat elliptical path. The ANOVA revealed the difference between groups to be highly significant, $F(2, 74) = 57.80$, $p < .01$, $\eta_p^2 = .62$. The variation across

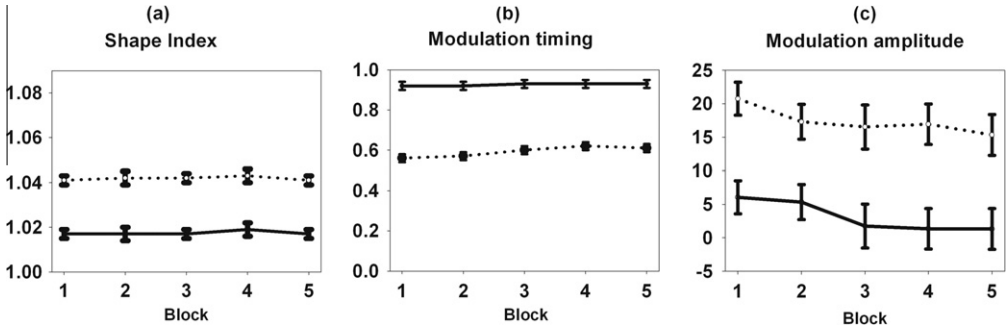

Fig. 4. Shape index (a), modulation timing (b) and modulation amplitude (c) for the experimental (continuous line) and the control group (dotted line) across the 5 blocks during the practice phase. Vertical bars represent standard errors.

blocks, $F(4, 74) = 0.65$, $p = .56$, $\varepsilon = .64$, $\eta_p^2 = .02$, as well as the interaction, $F(2, 74) = 0.09$, $p = .95$, $\varepsilon = .64$, $\eta_p^2 = .00$, were not significant.

In the experimental group the timing of the modulation of resultant velocity was highly regular. The mean modulation timing as measured by the cross-correlation was $M = .93$ ($SE = .02$) and did not vary across blocks. In the control group the mean modulation timing was only $M = .56$ ($SE = .02$) in the first block of trials and reached $M = .62$ ($SE = .02$) in the fourth block (in the fifth block it was $M = .61$, $SE = .02$). The difference between groups was significant, $F(2, 74) = 151.19$, $p < .01$, $\eta_p^2 = .81$. The main effect of block, $F(2, 74) = 5.56$, $p < .01$, $\varepsilon = .66$, $\eta_p^2 = .13$, and the interaction, $F(2, 74) = 3.14$, $p = .03$, $\varepsilon = .66$, $\eta_p^2 = .08$, also reached statistical significance.

In the experimental group the mean modulation amplitude of resultant velocity was $M = 3.16$ mm/s ($SE = 2.53$ mm/s), that is, the amplitude of the modulation was only marginally stronger than required. The exaggeration of the modulation was more pronounced in the control group, for which the mean modulation amplitude was $M = 17.36$ mm/s ($SE = 2.53$ mm/s). The difference between groups was significant, $F(2, 74) = 15.74$, $p < .01$, $\eta_p^2 = .30$, and also the main effect of block reached statistical significance, $F(2, 74) = 3.35$, $p = .05$, $\varepsilon = .41$, $\eta_p^2 = .08$, whereas there was no significant interaction, $F(2, 74) = 0.36$, $p = .65$, $\varepsilon = .41$, $\eta_p^2 = .01$.

3.2. Test phase

The means of the three dependent variables in the test phase are shown in Fig. 5. Both groups produced somewhat elliptic trajectories. The mean circularity indices were $M = 1.065$ ($SE = .004$) for the experimental group and $M = 1.072$ ($SE = .004$) for the control group. The main effect of group was

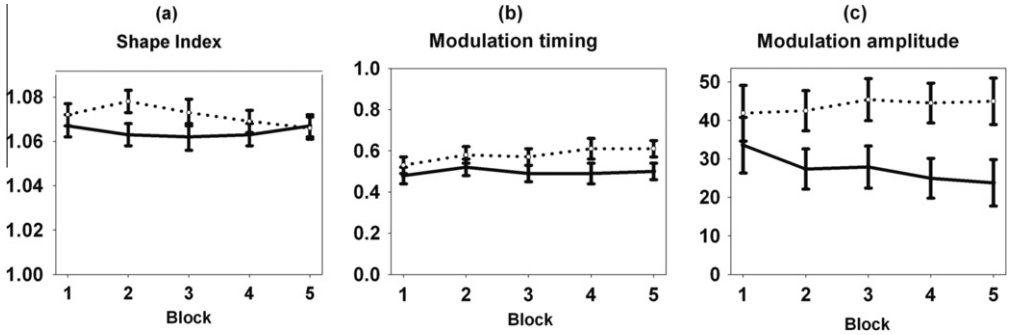

Fig. 5. Shape index (a), modulation timing (b) and modulation amplitude (c) for the experimental (continuous line) and the control group (dotted line) across the 5 blocks during the test phase. Vertical bars represent standard errors.

not significant, $F(2, 74) = 1.38$, $p = .25$, $\eta_p^2 = .04$. Also the main effect of block, $F(2, 74) = 0.61$; $p = .59$, $\varepsilon = .68$, $\eta_p^2 = .02$, and the interaction, $F(2, 74) = 1.48$, $p = .23$, $\varepsilon = .68$, $\eta_p^2 = .04$, failed to reach statistical significance.

The modulation timing of the resultant velocity was similar in both groups, but somewhat higher in the control group. In the experimental group the mean modulation timing was $M = .50$ ($SE = .04$) and in the control group it was $M = .58$ ($SE = .03$). According to the ANOVA the main effect of group fell short of statistical significance, $F(2, 74) = 2.37$, $p = .13$, $\eta_p^2 = .06$, and so did the main effect of block, $F(2, 74) = 2.97$, $p = .06$, $\varepsilon = .52$, $\eta_p^2 = .08$, and the interaction, $F(2, 74) = 1.34$, $p = .27$, $\varepsilon = .52$, $\eta_p^2 = .04$.

Both groups exaggerated the amplitude of the modulation of the resultant velocity, that is, their modulation amplitude was higher than required. As during practice, this exaggeration was more pronounced in the control group ($M = 43.85$ mm/s, $SE = 5.22$ mm/s) than in the experimental group ($M = 27.51$ mm/s, $SE = 5.22$ mm/s). This difference was significant, $F(2, 74) = 4.89$, $p = .03$, $\eta_p^2 = .12$. Whereas the modulation amplitude was consistently high across blocks of test trials in the control group, it declined slightly in the experimental group from a mean of $M = 33.56$ mm/s ($SE = 7.24$ mm/s) in the first block to a mean of $M = 23.76$ mm/s ($SE = 6.03$ mm/s) in the last block. However, neither the main effect of block, $F(2, 74) = 0.45$, $p = .70$, $\varepsilon = .69$, $\eta_p^2 = .01$, nor the interaction, $F(2, 74) = 1.40$, $p = .25$, $\varepsilon = .70$, $\eta_p^2 = .04$, reached statistical significance.

4. Discussion

The present study had been designed to examine the effects of robot assistance on performance and learning of an untypical velocity profile in drawing circles. With respect to the immediate performance effects, our findings are consistent with previous observations that robotic guidance serves to enhance performance and to reduce errors (Marchal-Crespo & Reinkensmeyer, 2008; Reinkensmeyer & Patton, 2009). In the experimental group, who practiced with robot guidance, the trajectories were almost perfectly circular and the velocity profile matched the desired profile quite well with respect to its timing and its modulation amplitude. In contrast, in the control group, who practiced without robot guidance, the trajectories were more elliptic and the modulation timing was less regular. Though there was an improvement across blocks regarding the modulation timing, this group performed the tracking task rather poorly and had difficulties in following the target, perhaps because of its modulated velocity. In addition the modulation amplitude, more precisely, the excess over the correct modulation amplitude, was much stronger than in the experimental group. Both groups showed an improvement in the modulation amplitude across blocks. Thus, all in all the movements experienced by the experimental group during practice were much more accurate than the movements experienced by the control group, but the accurate movements resulted primarily from external forces, generated by the robot, whereas the less accurate movements resulted from self-generated forces.

In the test trials, when robot guidance was switched off in the experimental group, the performance differences between groups largely disappeared. The accurate performance experienced with robot assistance could not be maintained in general. Only one of the differences present during practice was still present in test trials, though it was reduced. This was the difference in the amplitude of the velocity modulation which remained more accurate in the experimental group than in the control group. This difference between groups was not associated with a corresponding difference in the circularity index. Thus it reflects a certain degree of dissociation of curvature and velocity. Because there were no significant differences in modulation timing between groups, our results do not fully support the claim that learning of temporal aspects is qualified to be supported by robotic guidance (Feygin et al., 2002; Marchal-Crespo & Reinkensmeyer, 2008; Milot et al., 2010). Instead the benefits are likely to be limited to certain temporal characteristics.

The finding that robot assistance did not enhance learning of the timing of the velocity modulation, but only of its amplitude, can be taken to suggest that it was caused by a secondary effect of robot guidance. In the experimental group with robot guidance the cursor and the target were almost perfectly aligned throughout the whole practice phase. Therefore velocity modulations were not as obvious in the visual display as in the control group, where the target and the cursor drifted apart when the target was accelerated. Whereas participants in the experimental group modulated the velocity

fairly accurately in amplitude, participants in the experimental group learned to rush at points of high velocity and transferred this exaggeration into the test phase. The exaggeration of the velocity modulation in addition could have enhanced the modulation timing in the control group. Participants in the control group could have paid more attention to the more conspicuous velocity changes, resulting in better learning. Several authors (e.g., Connolly & Jones, 1970; Rock & Victor, 1964) have argued that individuals rely more on visual than on kinesthetic feedback. Thus, in the present study the visual indications of the velocity modulation might have been more important for learning than the kinesthetic information, and the visual information was less pronounced in the experimental group than in the control group. Thus the effects of robot guidance per se may have been superposed by the effects of robot guidance on the visual information during practice.

Furthermore, the guidance in the experimental group might have evoked a passive attitude in that the participants did not actively try to track the target. Some authors argued that continuous robotic assistance might lead to decreasing effort, attention, and motivation (Marchal-Crespo & Reinkensmeyer, 2009). As also stated by Schmidt and Lee (1999), guidance prevents the participants from experiencing and correcting errors and could thus be obstructive for the learning of movements. Therefore assistance that is adapted to the skill level of the participant or an assistance strategy that occasionally allows the participant to experience errors might be more beneficial.

In general performance was not very good in both groups. It is feasible that this is a result of the short duration of practice. The movement that was to be learned was a non-biological one, and participants might need several training sessions distributed over more than one day to learn it. The slightly diverging practice curves for the modulation timing and the modulation amplitude indicate that the asymptote had not yet been reached. On the other hand, the observation of practice effects after a single session is consistent with the findings of Beets et al. (2010). From the slight divergence of the practice curves one could expect that the difference between the two practice groups with respect to modulation amplitude could become more pronounced in the long run. In addition a superiority of the control group with respect to modulation timing could develop, but the present data do not allow a sufficiently firm conclusion in this respect.

As mentioned earlier by Reinkensmeyer, Galvez, Marchal-Crespo, Wolbrecht, and Bobrow (2007), the mechanisms of motor learning and the possible benefits by a robotic training are still not well understood. Additional investigations with other motor timing tasks as well as other assistance control strategies and a longer training period over several days shall further clarify the question if and under which conditions learning of temporal characteristics can be enhanced by robot assistance.

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